



Contract Playbook Comparison System

Stage 1 — Synthetic-Data Proof of Concept

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Why This Project

Manual contract review doesn't scale — and naive LLM use isn't safe



Slow, non-scaling review

Reviewers read 20–100+ page contracts manually, clause by clause.



Inconsistent outcomes

Different reviewers catch different deviations from the playbook.



No audit trail

Verdicts aren't linked back to the exact source text that produced them.



Hallucination risk

Feeding a whole contract to an LLM risks fabricated, unverifiable answers.



Our approach: a retrieval-augmented pipeline that grounds every verdict in an exact, cited span of the contract — never asking the LLM to judge from memory alone.

Validated market: Ironclad · LinkSquares · Lawgeex · Evisort (Contract Lifecycle Management)

What We Built at This Stage

A CPU-only proof of concept on a small, hand-authored synthetic dataset



Synthetic first

4 playbook clauses, 15 contracts —
full control of ground truth

CPU-only

No GPU dependency, no fine-
tuning, anywhere in the pipeline

Pluggable

Swapping in CUAD later = new data
loader, not a rewrite

Zero-API fallback

Rule-based comparator works with
no LLM access at all

How It's Built

Lightweight, CPU-friendly libraries behind a modular codebase



Tech Stack

- **Embeddings:** sentence-transformers — all-MiniLM-L6-v2
- **Vector index:** FAISS (CPU, IndexFlatIP)
- **Parsing:** pdfplumber (PDF + plain text)
- **Comparison:** LLM API, with rule-based fallback
- **Reporting:** fpdf2 (PDF) + CSV export
- **Demo UI:** Streamlit



Module Layout

ingestion/

PDF/text parsing, chunking, FAISS retrieval

comparison/

Rule-based + LLM comparator, evaluation

reporting/

CSV/PDF export, Streamlit app

common/

Shared schemas & config — the contract every module relies on

Finding & Fixing Real Retrieval Bugs

Every fix below was driven by measured evidence, not guesswork



01

Chunking was a no-op

BEFORE

Contracts fit inside a single 512-char chunk — every clause query retrieved the whole document.

AFTER

Reduced chunk size to ~120–200 chars, forcing real chunk boundaries.



02

Query/embedding mismatch

BEFORE

"Governing Law" queries were consistently outscored by unrelated boilerplate text.

AFTER

Rephrased the query to match legal register — Recall@3 jumped 57% → 100%.



03

Clause vs. chunk-size mismatch

BEFORE

Auto-Renewal's 150-char clause didn't fit inside a 120-char chunk — systematic truncation.

AFTER

Increased chunk size to fit the longest clause; all 4 clause types confirmed correct.

Retrieval Quality — Final Numbers

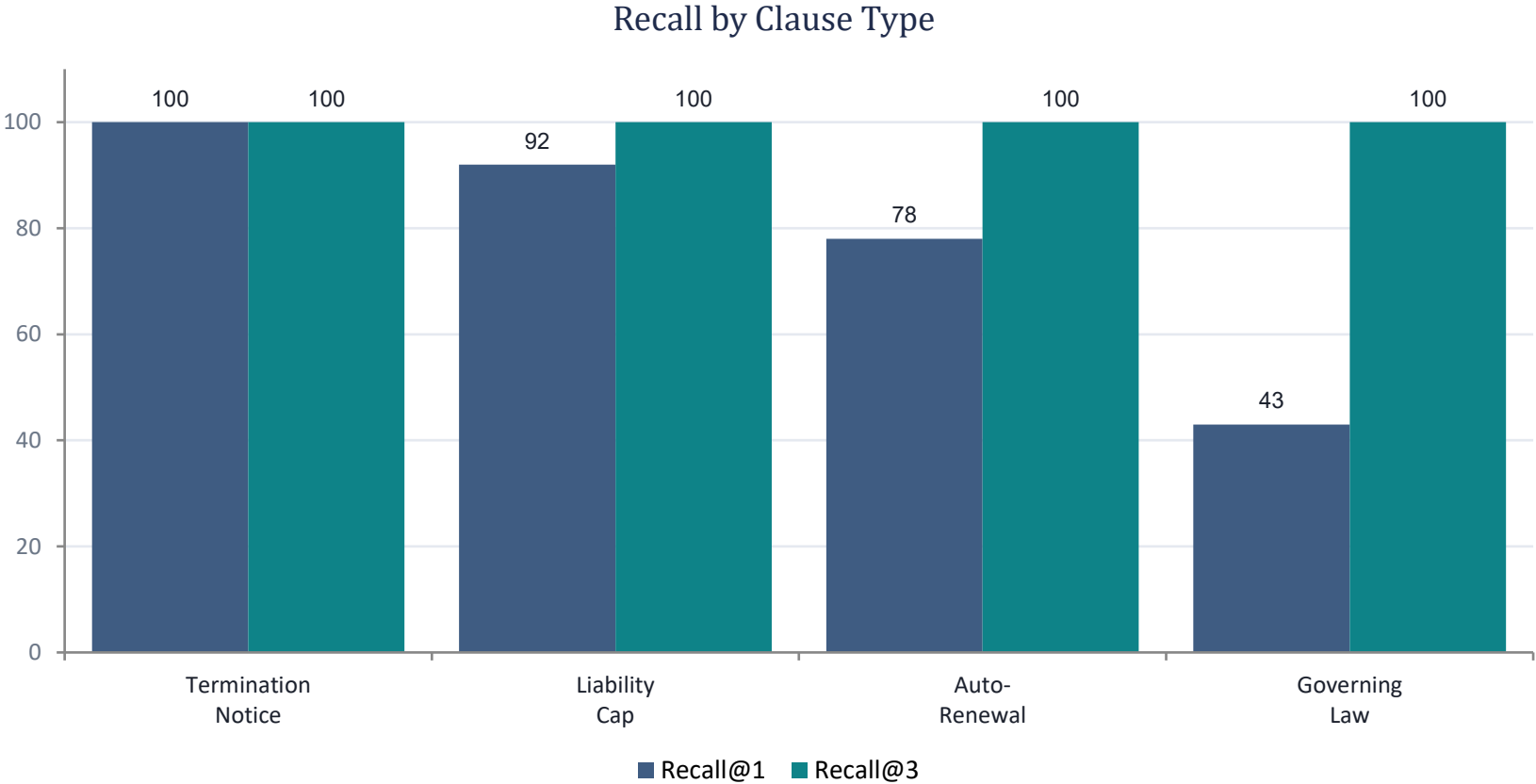
Recall@1 vs. Recall@3 across all four playbook clause types



100%

Recall@3

across all 4 clause types — the correct chunk is reliably in the retrieval shortlist.



Known limitation: the comparator currently uses only the top-ranked candidate — Governing Law's rank-1 precision (43%) trails its Recall@3 (100%), flagged for a Stage 2 fix.

What's Working End-to-End

Two runnable interfaces, a real test suite, and a three-branch team workflow



CLI Pipeline

main.py runs a contract end-to-end, outputs CSV + PDF report



Streamlit App

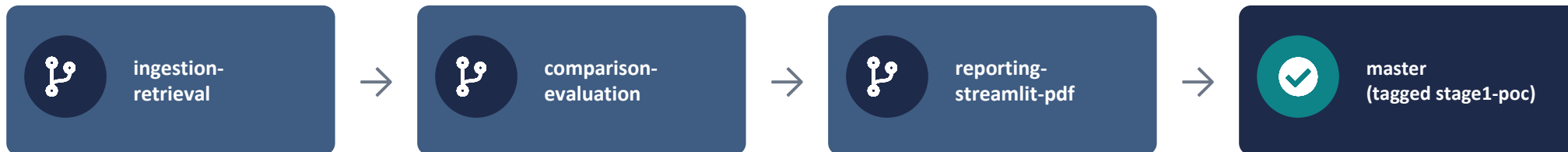
Upload or select a contract, view results, download reports



Test Suite

Full pytest coverage plus a real, measured evaluation script

Team Git Workflow



Each branch owns one pipeline layer, not one person's arbitrary slice — merged in dependency order: ingestion → comparison → reporting.

What's Next

Known limitations today, and the path to Stage 2



Known Limitations

- Comparator uses only the rank-1 candidate — fix identified, not yet applied
- Evaluated on synthetic data only — CUAD not yet integrated
- No hybrid BM25 retrieval yet — semantic-only for now



Stage 2 Roadmap

- Swap in CUAD via the existing pluggable data-source interface
- Add hybrid BM25 + semantic retrieval, with an ablation study
- Scale playbook from 4 to 5–8 clause categories
- Full Recall@k, MRR, and faithfulness evaluation per the proposal

Thank you